

Environmental pollution

What kinds of **pollution** are there in your surroundings? What are the main sources of pollution? How are you affected by pollution? Have you taken any step to reduce pollution and its harmful effects? What are the differences in **biodegradable** and **non-biodegradable** materials?

In this chapter, you will learn about:

- Pollution.
- Kinds of pollution. (Water, Air and Land).
- Main sources of pollution (smoke, sewage water, solid wastes, industrial wastes, etc.)
- Measures to reduce pollution.
- Biodegradable and non-biodegradable material.

All the students will be able to:

- ✓ Define pollution.
- ✓ Describe different kinds of pollution.
- ✓ Explain main causes of water, air and land pollution.
- ✓ Explain the effects of water, air and land pollution on the environment, and suggest ways to reduce them.
- ✓ Plan and conduct a campaign to bring awareness to the problem of environmental pollution in their surroundings.
- ✓ Differentiate between biodegradable and non-biodegradable material.
- ✓ Explain the impact of non-biodegradable material on the environment.
- Suggest ways to reduce the impact of non-biodegradable materials.



Figure 4.1: Environmental pollution

Kinds of pollution (Air, Land, Water)

Activity 1: Defining and describing kinds of pollution.

- 1 Define pollution.
- 2 Describe different kinds of pollution.

Do you know what pollution is? Look at the pictures below and name the kinds of pollution. Do you know what pollutants are? The pollutants are the harmful things that pollute the air, water and land environment, and make adverse changes to the air, water and land environment. Observe your surroundings. Talk to your elder siblings and peer. Draw or take and share photographs. Make a list of the pollutants in the air, water and land environment.



Teacher Note: Write the word “pollution” on the board or ask the students to observe the pictures above. Ask the students, “What is pollution?” What environments are polluted? Which pollutants can they observe in the pictures and in their living environments? Write student’s responses on the board.



Figure 4.2: Different kinds of pollution.

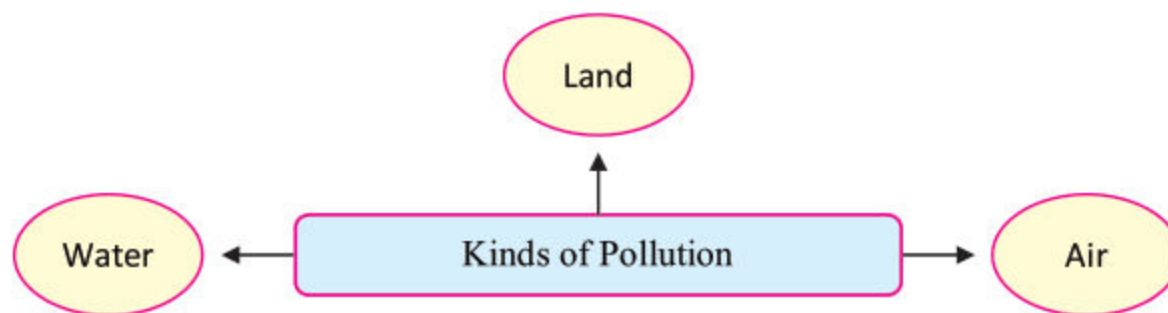
Many harmful substances called pollutants have been added to the environments causing **(Land, Water and Air) Pollution** in the above pictures. All living things in these environments have to experience the adverse effects of these changes to the environment. The air pollution is due to the release of smoke from the industries and vehicles into the clean air and the land pollution from dumping of solid waste and water pollution due to discharge of sewage from our homes and industries into the water bodies. **Smoke, solid waste and sewage** are **pollutants** that cause air, land and water pollution. These pollutants

are harmful for plants, animals and human beings living in these polluted environments.

What are the different polluted environments? What is pollution? What are pollutants?

Pollution is an undesirable change in the characteristics of air, water and land environment that is harmful to humans and other **living things** in the environment. Commonly, pollution is a form of addition of an undesirable and harmful substance like **smoke** or **non-biodegradable** material, like **plastic bags**, in the environment. Such **added substances** produced from human activities, bring about an undesirable change in the characteristics of air, water and land environment and are called **pollutants**.

Following are three types of pollutions:



Do you know?

Karachi is ranked at **fifth place** among the **top six polluted cities in the world**. The major cause of air pollution in Karachi is emissions from vehicle. In fact the residents suffer many health ailments due to the increasing rate of suspended particulates in the air. Failure to take appropriate steps to control the hazard could be very dangerous to the Lives of the people.

Main sources of pollution (smoke, sewage water and solid wastes)

1 Explain the main causes of water, air and land pollution.

We are affected by various kinds of environmental pollution in Pakistan. Look at the pictures. What are the main causes of these water, air and land pollution? Discuss with your class fellows and elder siblings and parents.



Figure 4.3: Air pollution

Air pollution is caused by the **solid, liquid** and **gaseous pollutants** released from the burning of fuel, such as coal, oil, natural gas, waste papers and trash in industries, vehicles and at homes. The **pollutant** is commonly known as smoke.



Figure 4.4: Water pollution

Water pollution is caused by the addition of **untreated sewage water** from our domestic use i.e., from washing and toilets, into the water bodies.

Water pollution is also caused by untreated water that contains various **harmful chemicals** from our industries.

Water pollution is also caused by the water run-off from agriculture that contains various **harmful pesticides and fertilizers**.



Figure 4.5: Land pollution

The **land pollution** is caused due to the dumping of solid wastes such as plastic bags, bottles, papers, glass bottles, tin cans, vegetable and fruit cuttings and other animal wastes and faeces on the land. These solid wastes that are dumped on to the ground are of two types. Do you know that some of these **solid wastes** are **biodegradable**; while others are **non-biodegradable**? Talk to your classfellows and elder siblings about biodegradable and non-biodegradable waste. Can you identify the biodegradable and non-biodegradable waste? Make a list of biodegradable and non-biodegradable waste that you throw away at home for one week. Share the list with your class fellows and siblings.

Cause of pollution

Talk to your class fellows and complete the table below:

Kinds of pollution	Causes

Effects of environmental pollution



Explain the effects of water, air and land pollution on the environment.

The prolonged exposure of human beings and living things to water, air and land pollution has harmful effects on their health. Read, share and explain the effects to your elder siblings and family member.

Kinds of pollution	Effects
Air pollution	Headache , eyes and nose irritation, sore throat, nausea, ill feeling, asthma, drowsiness, weakened judgement and lung diseases, such as pneumonia and cancer.
Water pollution	Cholera, typhoid , diarrhoea and other water borne diseases mostly related to digestive tract problems in human beings. Death of fishes, aquatic animals and plants living in the water or sea shores. Immune depression, reproductive failure or acute poisoning.
Land pollution	Toxic substances from the decay of land pollutants effect the underground water and the surrounding air.

Measures to reduce pollution

 Suggest ways to reduce environmental pollution.

Environmental pollution is mainly caused by human activities and it could be reduced by taking the following measures at individual and collective levels.

- **Reduce** waste by refusing the use of unnecessary materials and non-biodegradable waste.
- **Reuse** and **recycle** non-toxic cans, bottles, pots and other pottery, plastic, metal, glassware and paper.
- Use **public transport** or carpools.
- Do not burn **trash** and **waste**.
- Do not **dispose** solid waste in coastal areas or water bodies.
- Use **eco-friendly fuels**, washing liquids and bathroom cleaners.
- **Recycle** water used at home for cleaning purposes.

Activity 2: Awareness campaign on problems of environmental pollution in our surroundings.

 Plan and conduct a campaign to bring awareness to the problem of environmental pollution in one's surroundings.

What I need:

Observe in your surroundings.

- Talk to your pair, family members, elders, teachers, peer and senior fellows.
- Visit a library, visit the **IUCN** website or other websites and watch TV.



What to do:

1. Your teacher will form groups of four or five students in a team.
2. Ask each member to observe and record problems of environmental pollution in his or her surroundings.
3. Draw or bring a picture/photograph of three major environmental problems from your own surrounding.
4. Along with the teacher select one or two major problems faced by the community and explore its causes and effects. Make a poster.
5. Discuss and share the poster on the environmental problem, its causes, effects and ways to reduce it at the school assembly and in an event with parents and community members to create an awareness.
6. Ask all to sign a petition in order to take a personal action towards reducing environmental pollution.

I will take actions to reduce environmental pollution.	Name and signature

Teacher Note: Engage students in the activity and help them write the causes, effects and ways to reduce pollution. Engage the students in developing a poster and presentation skills.

What I observed: (remember to write your team member names)

Team members	Kind of pollution /Picture of Environmental Pollution	Causes/Effects

Activity questions:

1. What were the effects of the environmental pollution? Whom did you consult?
What did you read?
2. What were the most common causes of environmental pollution?
3. Which actions could be taken to reduce environmental pollution?

Do you know?

Microbial pollutants from sewage water are a major problem in the developing world that causes diseases such as **cholera, diarrhoea** and **typhoid** being the primary cause of children death in our community.

Biodegradable and non-biodegradable materials:

Differentiate between biodegradable and non-biodegradable materials

- 1 Differentiate between biodegradable and non-biodegradable materials.
- 1 Explain the impact of non-biodegradable material on the environment.
- 1 Suggest ways to reduce the impact of non-biodegradable materials.

We have various types of solid wastes. Some of the solid waste can be easily reduced through the natural waste reduction processes. These solid wastes that are easily decomposed by decomposers are called biodegradable. Other solid wastes that are not easily decomposed by decomposers or microorganisms, and that remain undecomposed on the earth for many years are called non-biodegradable wastes. Observe the picture A and B. What is similar and different materials?



Picture A



Picture B

Differences A	Similarities	Differences B

Major effects of non-biodegradable waste on the environment:

- Pollute ground water by emitting toxic pollutants into the ground water.
- Pollute air by emitting toxic gases on heating and burning.
- Polystyrenes and plastics can get into the oceans and water bodies and effect aquatic animals.
- Polystyrene and plastics break down into smaller pieces in the seas and oceans, causing small fishes and birds to accidentally swallow these pieces, thus endangering their lives.
- These wastes also choke the drainage systems and lead to over-flowing of gutters and flooding.

Figure 4.6: Effect of non-biodegradable waste on marine life



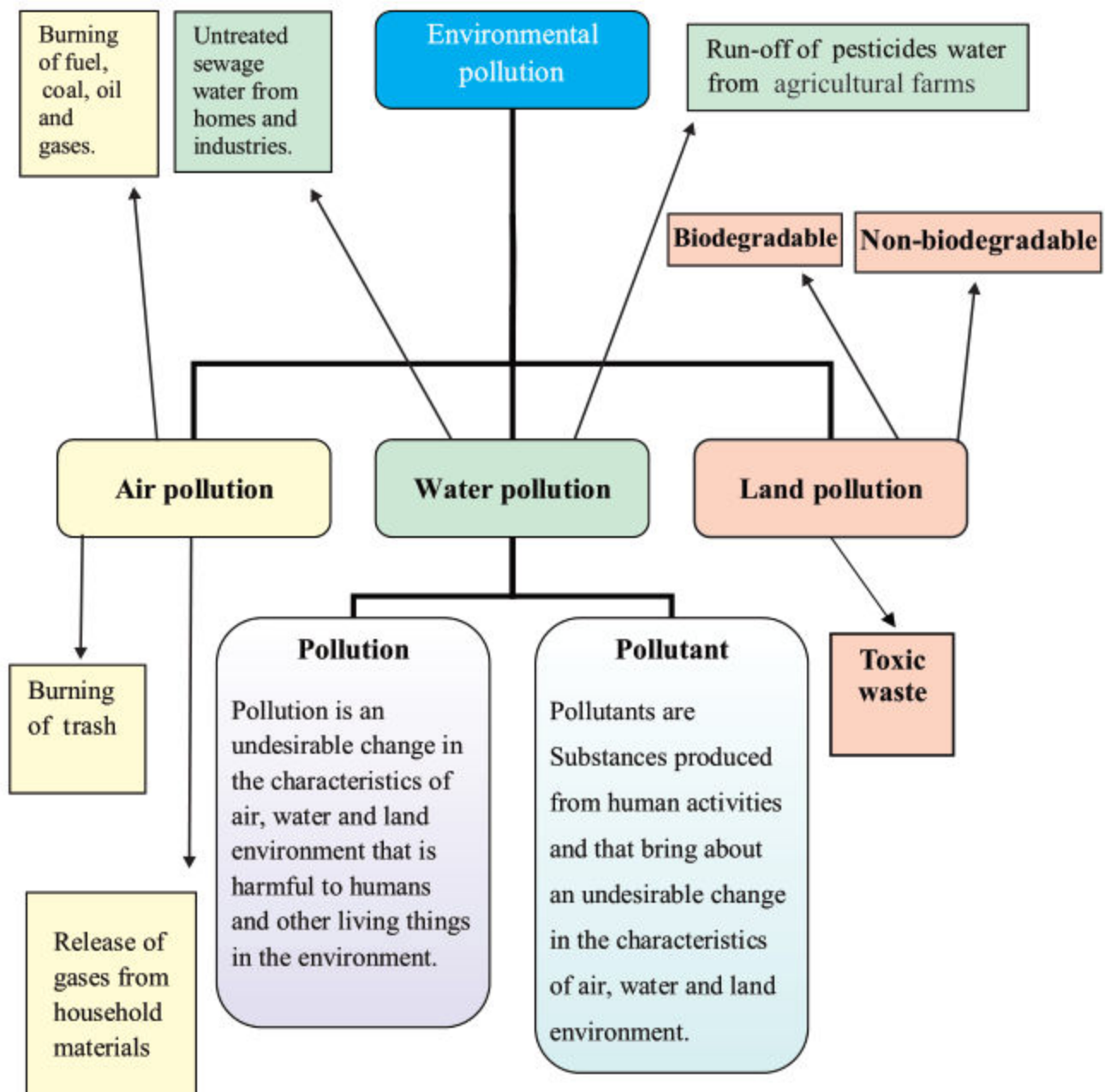
You have studied that solid wastes have several adverse effects on our environment.

According to the United Nations Environment Program plastic is killing a million seabirds a year, year, and 100,000 marine mammals and turtles worldwide. It kills by entanglement, most commonly in discarded synthetic fishing lines and nets. It kills by choking throats and gullets and clogging up digestive tracts, leading to fatal constipation in marine life. Bottle caps, pocket combs, cigarette lighters, tampon applicators, cotton bud shafts, toothbrushes, toys, syringes and plastic shopping bags are routinely found in the stomachs of dead seabirds and turtles.

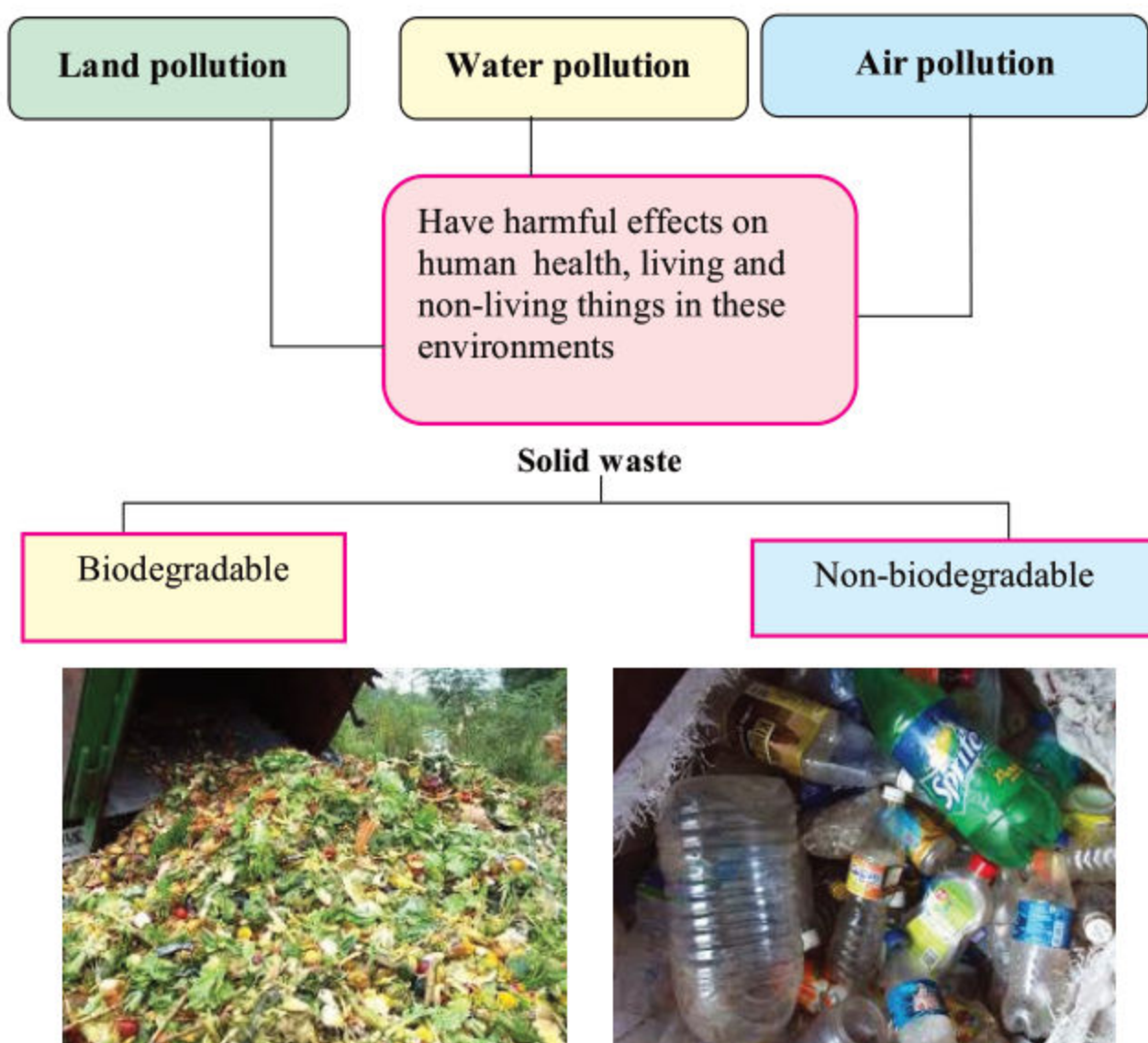
So to avoid such an we should take the following steps:

- 1) Avoid the use of non-biodegradable material.
- 2) Recycle the used materials for storing non-eatables.
- 3) Avoid heating and burning of plastics and polystyrene.
- 4) Do not throw away debris into the water bodies and open places.

Summary



Kinds of Pollution



Activity questions:

1. Circle T for True and F for False statements:

- | | | |
|----|--|-----|
| a. | Air pollution is caused by burning of fuels. | T F |
| b. | Water pollution is caused by burning of trash. | T F |
| c. | Land pollution is caused by disposing solid waste. | T F |
| d. | Non-biodegradable wastes do not harm marine life. | T F |
| e. | Drinking of polluted water causes water borne diseases. | T F |
| f. | Effects of breathing polluted air are headaches, eyes and nose irritation. | T F |
| g. | Agricultural run-offs pollute the water bodies. | T F |

2. Circle the best answer

- (i) What is **not** a cause of air pollution?
- a. burning of fuel b. burning of solid waste c. water from domestic use
- (ii) Which is **not** a biodegradable waste?
- a. vegetable peeling b. plastic bottles c. Paper
- (iii) Which of these are **not** a cause of water pollution?
- a. Gases from household materials
- b. Sewage water from homes
- c. Untreated water from industries
- (iv) If a child has cholera, which environment pollution needs to be addressed?
- a. Land b. Water c. Air
- (v) Asthma is due to the exposure to a type of environmental pollution?
Which one?
- a. Air b. Water c. Land
- (vi) What is an undesirable change in the characteristics of air, water and land that effects human health?
- a. Harmful b. Pollutant c. Pollution
- (vii) Which is the substance that causes undesirable change to the environment?
- a. Pollutant b. Harmful c. Pollution
- (viii) What is the major source of realease of direct pollutants in air pollution?
- a. Air b. Fuel c. Gas

Use the following picture to answer the two questions that follow.



(ix) Which type of wastes are classified as biodegradable waste?

- a. Waste from plastic industries.
- b. Vegetable waste from farms.
- c. Wasted water from homes.

(x) Which wastes are classified as non-biodegradable waste?

- a. Waste from the plastic industry.
- b. Vegetable waste from farms.
- c. Kitchen waste from home.

3. Give reasons

Air pollution can be reduced by not burning trash.

Plant and animal life can be protected by avoiding the use of plastics.